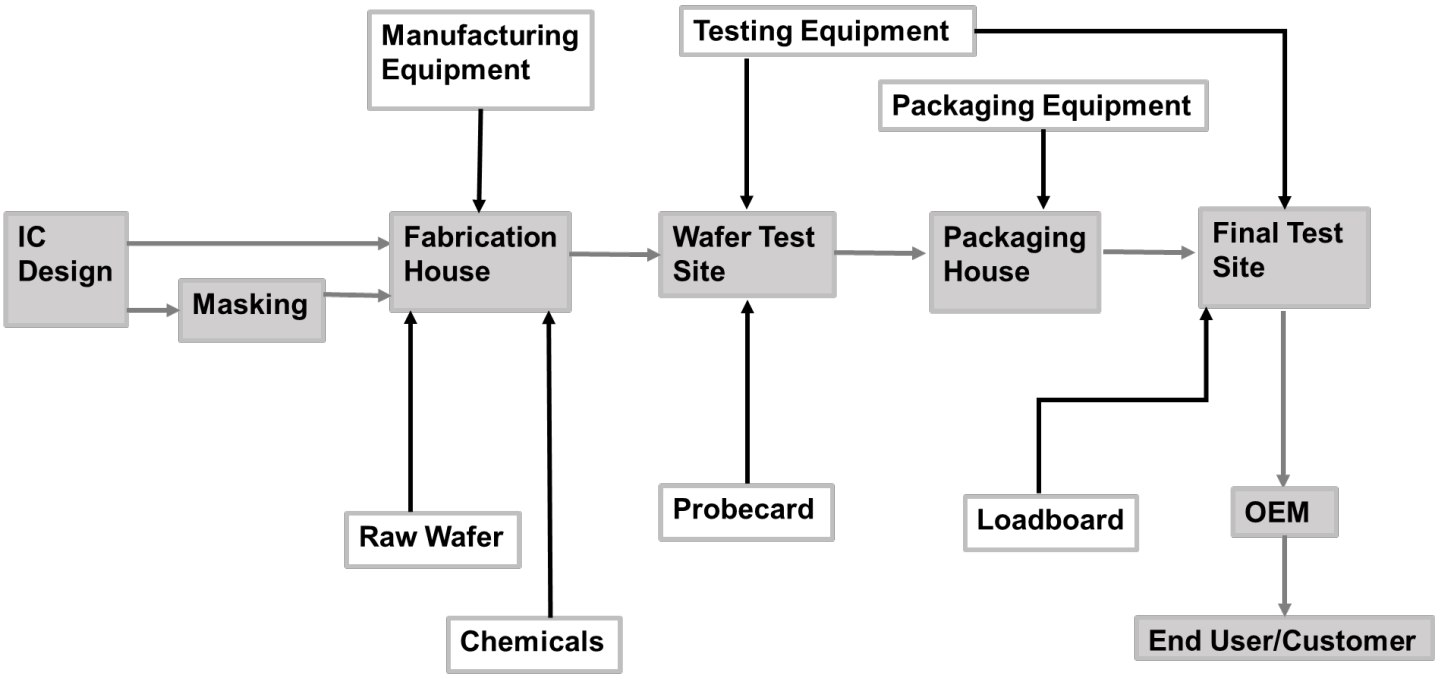
Application of Robust Optimization techniques to Wafer Supply Chain

Problem Overview: This study is inspired by the unique supply chain problems faced by 'fabless' semiconductor device manufacturing companies such as Qualcomm (where I used to work previously). The companies following the fabless model, by definition, have no fabrication/manufacturing plant of their own. Hence, they have to use complex supply chain strategies to source, fabricate, assemble, and package their integrated chips.

A basic schematic of the semiconductor lifecycle is shown below:



Briefly, a typically integrated chip goes through the following steps in chronological order:

1. The company specifies the chip's design and transmits it to the fabrication house (companies like TSMC, GlobalFoundries, and Samsung are big players).
2. The fabrication house uses lithographical techniques on silicon substrate wafers to produce individual integrated chips called dies.
3. A finished wafer has both good chips and bad chips, and the ratio of good chips to the total is referred to as its yield.
    a. Some unique strategies are possible to 'harvest' or find a use for 'partially' good dies. For example, a die may have a functional camera block but a non-functional modem. It is possible to find customers that may use these chips, but the return is doubtful.
    b. The classification of good vs. partially good vs. bad dies is done via test programs on specialized equipment known as Automated Test Equipment (or ATE). Some common vendors for this include Advantest and Teradyne. Additionally, specialized hardware (colloquially called 'probe card') is needed to carry this out. This stage of the testing process is commonly called 'Wafer Sort.'
    c. Broadly, Testing is divided into two categories – structural and performance/functional. Structural tests can be simplified as discrete 'connectivity' tests, while performance tests ensure that the minimum functional requirements are met.
4. The confirmed good chips from wafer testing are then passed on to the Assembly and Packaging (SPIL, ASE, and Amkor are examples of such companies).
    a. This manufacturing process stage involves transferring the good wafer die into a package (an easy way to think of plastic housing).

b. These 'finished goods' are then passed through the second round of Testing, a stage commonly called 'Final Test.' Testing at this stage is done at a different temperature from the initial Wafer Sort. Typically, yields here are much higher since the parts have already been tested once. The equipment used for this step is the same as Wafer Sort, but the hardware is different and is colloquially called 'load boards.'
c. Following Final Test, parts are put through another testing process at the system level, where the chip is put through tests that verify whether it can work in concert with other chips. For example, the key IC components of a mobile phone are MSM, PMIC, and RF chips. System-Level Testing verifies that all of these talk to each other and work as expected.
d. A final sanity check is done via sampling, a process known as Quality Assurance.

Delving further into Step 3 in the process steps outlined above, there are two strategies open to a fabless company:

1. Known Good Die or KGD model: Pay a fixed fee per confirmed good die in the wafer as determined by the Wafer Sort program created by the company. The fabrication plant takes on all the responsibility for logistics of the wafer testing, transportation, yield analysis, process tuning, etc. Any material loss after good dies is accepted and borne by the company.
2. Wafer Buy or WB model: Pay a fixed fee per wafer. The company takes on all responsibility for the wafer testing logistics, transportation, yield analysis, process tuning, etc. Since the company owns the wafer, some creative structures can be built to take advantage of 'partially good' parts, as described earlier. Such a structure is not possible with the KGD model.

Constraints: The supply strategy will minimize the per-good die cost. However, there exist limitations:

1. Capacity constraint – One supplier often does not have enough capacity for the required throughput. Multiple fabrication houses may be required, and each one has its own cost and quality characteristic. Additionally, there will be a limited capacity for WB model wafers as the fabrication house will prefer to use the KGD model to utilize their expertise better and improve their revenue.
2. Throughput and Demand constraint – A specific throughput of good parts needs to be planned to possess enough inventory to meet or exceed demand.
3. Hardware time constraint – In both the WB and KGD models, the time windows for Testing will be limited by the test sites and hardware available for use. Hence, the test-time per wafer becomes a constraint that needs to be met for throughput.

Uncertainties: Several sources of uncertainty exist in such a complex supply chain:

1. Yield uncertainty – Wafer yields tend to follow a normal distribution per fabrication house. With a multiple-sourcing strategy and production deviations, there will be uncertainty in output.
2. Hardware time uncertainty – Related to wafer yields, the relative time to test each wafer will vary depending on its quality. If a wafer has a lower yield with a similar time taken, it reduces throughput.
3. Uncertainties downstream – Some cost uncertainties in a) Shipment, b) Inventory storage c) Process and yield analysis – are considered.
4. WB' partially good' die revenue – The magnitude of the revenue streams from these dies are uncertain.

Benefits of Robust Optimization approach: The semiconductor supply chain has many points at which disruptions can occur. This is something I've seen personally during the seven years I worked at Qualcomm, where chip shortages led to a significant loss of revenue and an intangible cost in terms of customer trust on one of the company's very high volume chips. The uncertainties listed above are not exhaustive, but a robust optimization approach to protecting against these uncertainties will undoubtedly be beneficial.

Related Work: This is an interesting case of a sourcing/supply chain optimization problem, where the cost needs to be minimized while maximizing revenue. Some related work has been done in the areas of production planning and lot allocation by Ng et al. [1 and 2], on capacity planning by Barahona et al. [3], and on yield uncertainty planning by Lowe et al. [4]. However, there has been no attempt to model the entire process, which might pose an interesting, albeit much larger, and more complex, problem. The current project applies the assumptions covered in the next section to simplify the problem at hand somewhat. It essentially treats the latter portion of the semiconductor lifecycle with a scaling factor left uncertain in the robust formulation.

Implementation: To frame the problem, some simplifying assumptions are made:

1. Two wafer suppliers, A and B, are considered, with fixed capacities for KGD and WB wafers.
    a. A has higher quality, more tightly controlled process metrics, and higher cost.
    b. B is cheaper, lower quality, shows a much broader process variation, and has a higher capacity to commit to the company both in KGD and WB.
    c. As a proxy for these factors, the robust model uncertainty sets were modified to allow a wider variation for B than A and included a higher baseline yield for A.
2. KGD wafers per-good die cost was fixed since these are generally bound by a contract.
3. WB wafers per-wafer cost was fixed since these are generally bound by a contract.
4. This problem assumes that finished goods are available in inventory in advance of demand hitting. This is reasonable since OEMs do submit buy orders with a lead time of at least one quarter.
5. The problem assumes operations for one quarter, and different demand levels are modeled (in millions: 1, 5, 10, 20, 30, 40, 50) for comparison.
6. The wafer capacities are such that the maximum 50 million demand can be met by the two suppliers combined, but not individually. The ratio of WB to KGD capacities for each supplier is A – 3:7 and B – 2:3. Additionally, the total maximum wafer capacity of A to B is 1:2.
7. The extra costs of WB in terms of transportation and testing/engineering support are calculated per wafer, while the additional hardware costs are in terms of pieces of hardware that require fabrication.
8. The downstream material loss (from the final test, system-level test, and quality assurance) is modeled via a scaling factor that is higher for A and lower for B.
9. Revenue is calculated as demand multiplied by sale price to the OEM, plus any additional proceeds from the WB' partially good' parts. This is nominally set to 2.5% of the total value of the good parts in the wafer.
10. Downstream cost is modeled as a numeric value combining all downstream processes and is uncertain.
11. Total cost is simply the sum of all KGD, WB, and downstream costs.

All the values used are notional and synthetically generated but indicative based on previous experience within the industry. No publicly available datasets were found or used in the problem.

Observations: The model runs showed the following observations:

1. Since the WB model has a fixed cost, lower demand levels only recommend using the KGD model. This is consistent with the general real-world observations.
2. Robustness adds a cost of 13-16% as we go from the highest demand level to the lowest. This is consistent with the fact that as we add the WB mode into our mix, the average cost of each new unit is lower.
3. Robustness is seen to cost the program with a profit (revenue minus total cost) reduction of 27-28%.

Conclusion: Based on our study of the nominal and robust model recommendations at different levels of demand using synthetically generated numbers for each uncertain variable with differing distributions, we conclude:

1. Robustness ensures that demand is served in all scenarios. Our current safety factors lead to a surplus of 4-18% in the simulated cases.
2. In the nominal solution, there is a deficit in the demand in some cases, and it goes up to 8%.

Due to market dynamics and other factors, semiconductor ICs generally have a ’long tail’ in their product lifecycle. The same chips may be sold in smaller volumes for up to a decade after the initial high-volume product ramp. Hence, the surplus seen in the robust case can be used to fill buy orders later in the future, though there may be further degradation and material losses due to storage and other factors. That is to say, some of the ‘costs of robustness’ could be recouped in this fashion.

Citations:

[1] ‘Semiconductor production planning using robust optimization,’ T. S. Ng, J. Fowler, 2007 IEEE International Conference on Industrial Engineering and Engineering Management

[2] ‘Semiconductor lot allocation using robust optimization,’ T. S. Ng, Yang Sun, J. Fowler, European Journal of Operational Research, 21 January 2010

[3] ‘Robust capacity planning in semiconductor manufacturing,’ Francisco Barahona, Stuart Bermon, Oktay Günlük, Sarah Hood, Naval Research Logistics, Vol. 52 (2005)

[4] ‘Robust semiconductor production planning under yield uncertainty, Jonathan J. Lowe, Amin Khademi, Scott J. Mason, IEEE 2016 Winter Simulation Conference (WSC)

Appendix:

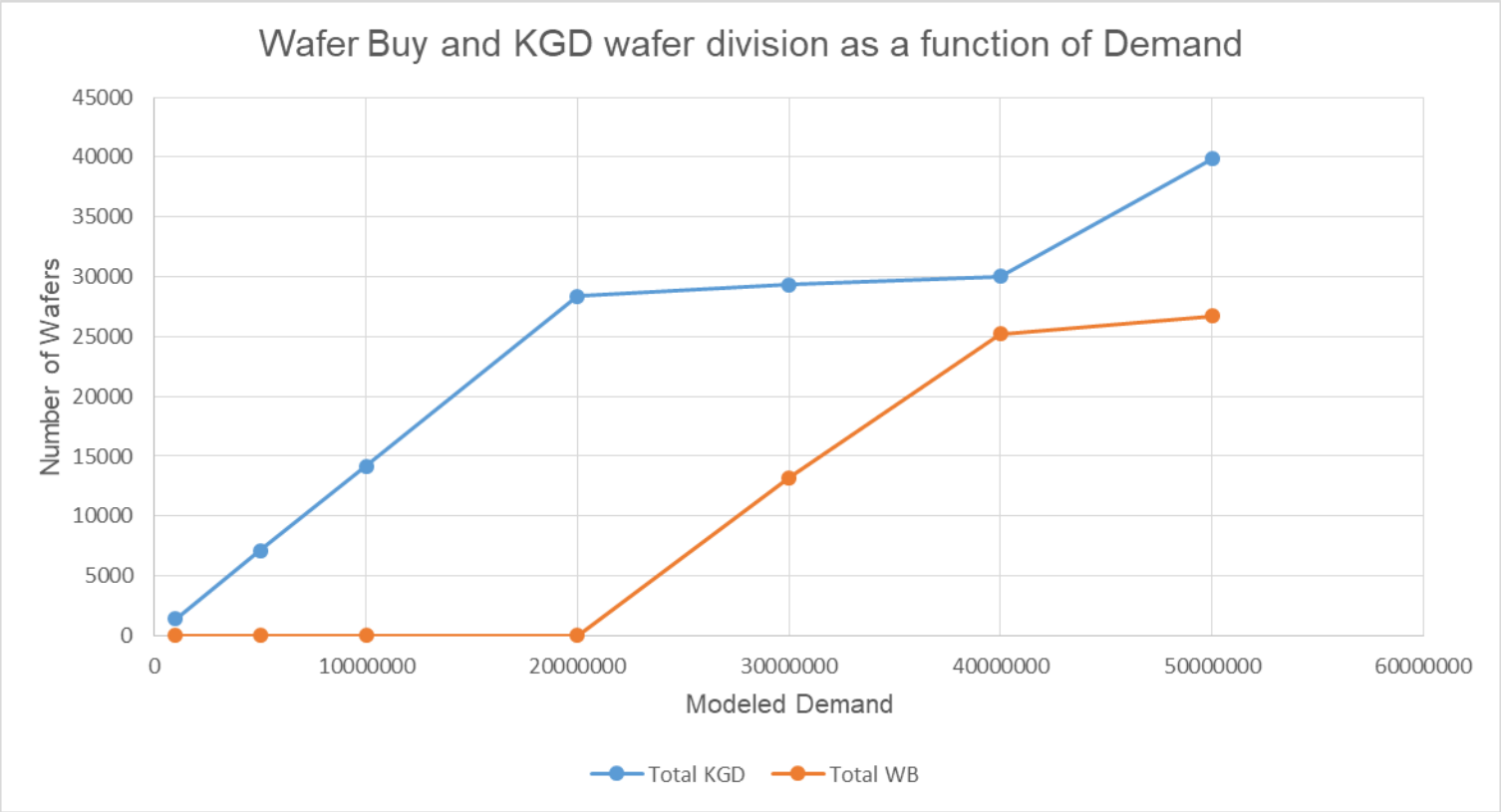
Problem formulation:

$$
\begin{aligned}
\max \quad & Revenue - TotalCost \\
\text{Subject to} \quad & WKGDa \geq 0 \\
& WKGDb \geq 0 \\
& WWBa \geq 0 \\
& WWBb \geq 0 \\
& -WKGDa \geq -17500 \\
& -WWBa \geq -7500 \\
& -WKGDb \geq -30000 \\
& -WWBb \geq -20000 \\
& CWB - 20000WWBa - 15000WWBb = 0 \\
& TestCostPerWafer = 2000 \\
& TransportCostPerWafer = 80 \\
& EngCostPerWafer = 5 \\
& HWPcsA - 0.0006666666666666666WWBa = 0 \\
& HWPcsB - 0.00083333333333333334WWBb = 0 \\
& Revenue - 656.25WWBb - 787.5WWBa = 3.5e7 \\
& \qquad - WWBa \times TestCostPerWafer - WWBb \times TestCostPerWafer - WWBa \times TransportCostPerWafer - WWBb \\
& \times TransportCostPerWafer - WWBa \times EngCostPerWafer - WWBb \times EngCostPerWafer + CWBextra - 300000HWPcs \\
& WKGDa, \in \mathbb{Z} \\
& WKGDb, \in \mathbb{Z} \\
& WWBa, \in \mathbb{Z} \\
& WWBb, \in \mathbb{Z} \\
& CWB \geq 0 \\
& CKGD \geq 0 \\
& TotalCost \geq 0 \\
& Revenue \geq 0 \\
& TestCostPerWafer \geq 0 \\
& TransportCostPerWafer \geq 0 \\
& CWBextra \geq 0 \\
& EngCostPerWafer \geq 0 \\
& HWPcsA, \in \mathbb{Z} \\
& HWPcsB, \in \mathbb{Z} \\
& DownstreamCost \geq 0 \\
& HWPcs, \in \mathbb{Z}
\end{aligned}
$$

Results:

Nominal											
Demand	TotalCost	Revenue	WaferBuy-A	WaferBuy-B	KGD-A	KGD-B	Wafer Buy Cost	KGD Cost	HW Pieces A	HW Pieces B	Downstream
1000000	22347000	35000000	0	0	2	1416	0	22347000	0	0	7000000
5000000	111705750	175000000	0	0	1	7091	0	111705750	0	0	35000000
10000000	223413750	350000000	0	0	0	14185	0	223413750	0	0	70000000
20000000	446811750	700000000	0	0	0	28369	0	446811750	0	0	140000000
30000000	691147500	1058662500	0	13200	0	29354	198000000	462325500	0	11	211732500
40000000	939604500	1417325000	6000	19200	25	30000	408000000	473062500	4	16	283465000
50000000	1194387000	1768506250	7500	19200	9863	30000	438000000	694417500	5	16	353701250

Robust											
Demand	TotalCost	Revenue	WaferBuy-A	WaferBuy-B	KGD-A	KGD-B	Wafer Buy Cost	KGD Cost	HW Pieces A	HW Pieces B	Downstream
1000000	25829775	35000000	0	0	2	1605	0	25829775	0	0	7070000
5000000	129130470	175000000	0	0	0	8038	0	129130470	0	0	35350000
10000000	258260940	350000000	0	0	0	16076	0	258260940	0	0	70700000
20000000	518727510	701181250	1500	0	130	29999	30000000	484888185	1	0	141638612
30000000	794876040	1061418750	7500	8400	0	29923	276000000	480712995	5	7	214406588
40000000	1073555910	1418506250	7500	19200	3937	30000	438000000	571418325	5	16	286538262
50000000	1350187335	1768506250	7500	19200	16110	30000	438000000	848049750	5	16	357238262



Results on synthetically adding uncertainty:

Robust			
Total Chips	**Demand**	**Deficit**	**Total/Demand**
1123677	1003000	-120677	1.120316052
5594448	4930000	-664448	1.134776471
10529780	10110000	-419780	1.041521266
23660399	20400000	-3260399	1.15982348
40426771	34230000	-6196771	1.181033333
47138277	43800000	-3338277	1.07621637
55067520	53350000	-1717520	1.03219344

Nominal			
Total Chips	**Demand**	**Deficit**	**Total/Demand**
991566	1003000	11434	0.988600199
4936191	4930000	-6191	1.001255781
9291175	10110000	818825	0.919008408
23858329	20400000	-3458329	1.169525931
37319858	34230000	-3089858	1.090267543
42153825	43800000	1646175	0.962416096
49870016	53350000	3479984	0.934770684